Compulsory Exercise

Topics in Life Insurance (Liv2), Block 3, 2026

March 18, 2026

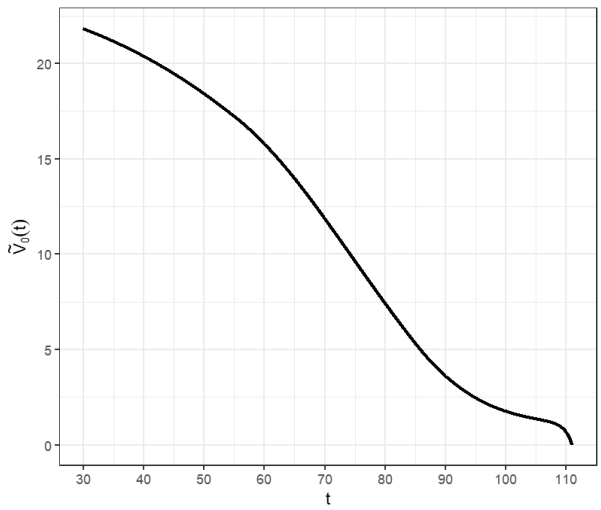
(i)

Using Lebniz' rule we obtain af ordinary backward differential equation for the reserve $\tilde{V}_0$. We have

$$\begin{aligned}
\frac{d}{dt}\tilde{V}_0(t) &= \int_t^\infty \frac{\partial}{\partial t} e^{-\int_t^s (\tilde{r}(u)+\tilde{\mu}(u))\,du}\,ds - 1 \\
&= \int_t^\infty e^{-\int_t^s (\tilde{r}(w)+\tilde{\mu}(w))\,dw}(\tilde{r}(t)+\tilde{\mu}(t))\,ds - 1 \\
&= (\tilde{r}(t)+\tilde{\mu}(t)) \int_t^\infty e^{-\int_t^s (\tilde{r}(w)+\tilde{\mu}(w))\,dw}\,ds - 1 \\
&= (\tilde{r}(t)+\tilde{\mu}(t))\tilde{V}_0(t) - 1,
\end{aligned}$$

with $\tilde{V}_0(111) = 0$.

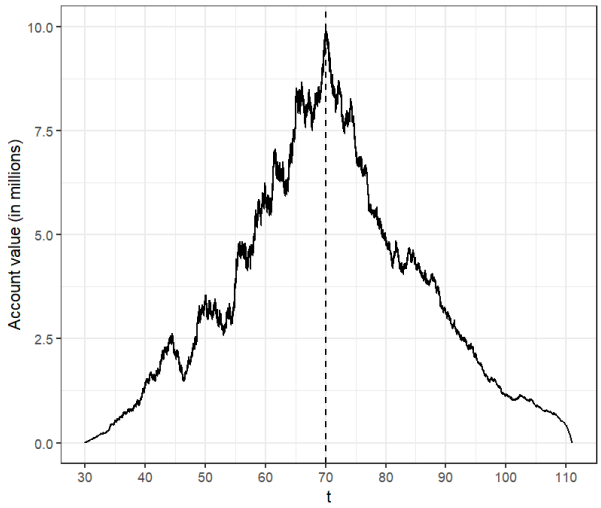
Now, using the fourth order Runge-Kutta method we optain a numerical solution of $\tilde{V}_0(t)$. The result is shown via the plot below



Exercise (ii)

(iii)

A simulation of the account is provided below.



We observe from our simulation that it begins from age $t = 30$ at the value of zero as demanded. Thereafter, the account accumulates until retirement $m = 70$ with a value of around 9.8 million (monetary units). Then the account de-cumulates until our assumed lifetime cut-off at $t = 111$. We also observe random fluctuates due to the fact that we model the risky asset via a Brownian motion.

Exercise (iv)

Assuming $V^0(t,x) = x$ and then showing that the two PDEs coincide we have shown $V^0(t,x) = x$.

From Proposition 3.3, assuming $V^0(t,x)$ is sufficiently differentiable and conditioning on $j = 0$, the market reserve satisfies the PDE

$$V_t^0(t,x) = rV^0(t,x) - b^0(t,x) - \mu(t)R^{01}(t,x) - \mathcal{D}_x V^0(t,x), \tag{1}$$

where $V^1(t,x) \equiv 0$, yielding

$$R^{01}(t,x) = b^{01}(t,x) - V^0(t,x),$$

and the operator $\mathcal{D}_x$ is given by

$$\mathcal{D}_x V^0(t,x) = V_x^0(t,x)(rx - a^0(t,x) - \mu(t)\varrho^{01}(t,x)) + \frac{1}{2}V_{xx}^0(t,x)\pi^2(t,x)\sigma^2 x^2,$$

Note in our case: $\varrho^{01}(t,x) = a^{01}(t,x) - x$.

Since the contract features no discrete jumps in benefits or account balance, the jump condition (3.7b) is satisfied.

For $V^0(t,x) = x$ we find the derivatives.

$$V_t^0 = 0, \qquad V_x^0 = 1, \qquad V_{xx}^0 = 0.$$

Substituting these into the operator yields $\mathcal{D}_x V^0(t,x) = rx - a^0(t,x) - \mu(t)\varrho^{01}(t,x)$ and $R^{01}(t,x) = b^{01}(t,x) - x$. The PDE (1) thus simplifies to checking whether the continuous risk-neutral account drift matches the benefits and mortality risks:

$$A(t,x) = rx - b^0(t,x) - \mu(t)\big(b^{01}(t,x) - x\big). \tag{2}$$

Active phase, $t \in [30, m)$

During the active phase, premiums are paid at rate $\theta L(t)$ (a negative benefit) and the death benefit is $KL(t)$. Thus, the actual contractual benefits are:

$$b^0(t,x) = -\theta L(t), \qquad b^{01}(t,x) = KL(t).$$

From SDE (1), substituting the physical expected return $\tilde{r}(t)$ with the risk-free rate r, the risk-neutral drift of the account is

$$A(t,x) = r\,x + \theta L(t) - \mu(t)\big(KL(t) - x\big).$$

We verify (2) by evaluating the right-hand side:

$$\begin{aligned} rx - b^0(t,x) - \mu(t)\big(b^{01}(t,x) - x\big) &= rx - \big(-\theta L(t)\big) - \mu(t)\big(KL(t) - x\big) \\ &= rx + \theta L(t) - \mu(t)\big(KL(t) - x\big) \\ &= A(t,x). \quad \checkmark \end{aligned}$$

Retirement phase, $t \in [m, 111]$

During retirement, the account pays a continuous annuity rate $x/\tilde{V}^0(t)$, and the account is forfeited to the mutual pool upon death (no death benefit). Thus:

$$b^0(t,x) = \frac{x}{\tilde{V}^0(t)}, \qquad b^{01}(t,x) = 0.$$

From SDE (2), the account growth includes a survivorship dividend $\mu(t)x$ from the mortality pool. Changing $\tilde{r}(t)$ to r, the risk-neutral drift is

$$A(t,x) = \big(r + \mu(t)\big)x - \frac{x}{\tilde{V}^0(t)}.$$

We verify (2) by evaluating the right-hand side:

$$\begin{aligned} rx - b^0(t,x) - \mu(t)\big(b^{01}(t,x) - x\big) &= rx - \frac{x}{\tilde{V}^0(t)} - \mu(t)\big(0 - x\big) \\ &= rx - \frac{x}{\tilde{V}^0(t)} + \mu(t)x \\ &= \big(r + \mu(t)\big)x - \frac{x}{\tilde{V}^0(t)} \\ &= A(t,x). \quad \checkmark \end{aligned}$$

Boundary condition

At the terminal age $n = 111$ there are no future obligations, so equation (3.7c) requires the boundary condition $V^0(111, x) = 0$. If $V^0(t,x) = x$, this means we need $X(111) = 0$.

This is guaranteed by the construction of the calculation basis: the reserve $\tilde{V}^0(t)$ computed in Exercise (i) satisfies $\tilde{V}^0(111) = 0$. Thus, the annuity payout rate $X(t)/\tilde{V}^0(t) \to \infty$ as $t \to 111$, which drains the account completely. The purpose of $\tilde{V}^0$ is precisely to calibrate the payout rate so that the account is exhausted at the maximum age. The simulation from Exercise (ii) confirms this numerically: $X(111) \approx 0$, thereby satisfying (3.7c).

Conclusion

Proposition 3.3 states that, conversely, if there exists a set of functions $V^j(t,x)$ such that conditions (3.7) hold, then the function is indeed equal to the reserve. By explicitly verifying that $V^0(t,x) = x$ satisfies the PDE (3.7a) in both phases, the trivial jump condition (3.7b), and the terminal condition (3.7c), we conclude that

$$\boxed{V^0(t,x) = x.}$$

The market reserve equals the account value. This is the defining property of a unit-link product: the policyholder bears all investment risk, and the account dynamics are constructed to continuously budget for every mortality and benefit cash flow, making the account the exact self-financing replicating portfolio for all future obligations.

Exercise (v)

For $t \geq m$, we can see from the account dynamics, $dX(t)$, that the life annuity benefit process is

$$Y(t) = b^0(t, X(t)) = \frac{X(t)}{\tilde{V}_0(t)}.$$

Since $\tilde{V}_0(t)$ is deterministic, we apply the product rule to $Y = X \cdot (1/\tilde{V}_0)$.

Ingredient 1: SDE for X

From SDE (2), for $t \geq m$:

$$\mathrm{d}X(t) = \left[(\pi(t)\alpha + (1-\pi(t))r + \mu(t))X(t) - \tfrac{X(t)}{\tilde{V}_0(t)}\right] \mathrm{d}t + \sigma\pi(t)X(t)\,\mathrm{d}W(t).$$

Ingredient 2: ODE for $\tilde{V}_0$

From the Thiele ODE derived in Exercise (i):

$$\tilde{V}_0'(t) = \big(\tilde{r}(t) + \tilde{\mu}(t)\big)\tilde{V}_0(t) - 1,$$

so that

$$\mathrm{d}\left(\frac{1}{\tilde{V}_0(t)}\right) = -\frac{\tilde{V}_0'(t)}{\tilde{V}_0(t)^2}\,\mathrm{d}t = \left[-\frac{\tilde{r}(t) + \tilde{\mu}(t)}{\tilde{V}_0(t)} + \frac{1}{\tilde{V}_0(t)^2}\right]\mathrm{d}t.$$

Product rule

Applying the product rule $\mathrm{d}Y = \frac{1}{\tilde{V}_0}\,\mathrm{d}X + X\,\mathrm{d}\big(\frac{1}{\tilde{V}_0}\big)$ (no cross-variation term since $\tilde{V}_0$ is deterministic), we get:

$$\frac{1}{\tilde{V}_0(t)}\,\mathrm{d}X(t) = \Big[(\pi(t)\alpha + (1-\pi(t))r + \mu(t))\,Y(t) - \frac{X(t)}{\tilde{V}_0(t)^2}\Big]\mathrm{d}t + \sigma\pi(t)\,Y(t)\,\mathrm{d}W(t),$$

$$X(t)\,\mathrm{d}\Big(\frac{1}{\tilde{V}_0(t)}\Big) = \Big[-(\tilde{r}(t) + \tilde{\mu}(t))\,Y(t) + \frac{X(t)}{\tilde{V}_0(t)^2}\Big]\mathrm{d}t.$$

The $X(t)/\tilde{V}_0(t)^2$ terms cancel, giving

$$\begin{aligned}
\mathrm{d}Y(t) &= \big(\pi(t)\alpha + (1-\pi(t))r - \tilde{r}(t) + \mu(t) - \tilde{\mu}(t)\big)\,Y(t)\,\mathrm{d}t + \sigma\pi(t)\,Y(t)\,\mathrm{d}W(t),\\
Y(m) &= \frac{X(m)}{\tilde{V}_0(m)}.
\end{aligned}$$

This is a (generalised) geometric Brownian motion since both the drift and diffusion coefficients are of the form $f(t)Y(t)$.

Interpretation of the calculation basis

The calculation basis $(\tilde{r}(t), \tilde{\mu}(t))$ determines the drift of $Y(t)$ and thereby the expected payout profile. With the specific choice from Table 2, $\tilde{r}(t) = \pi(t)\alpha + (1-\pi(t))r$ and $\tilde{\mu}(t) = \mu(t)$, the drift vanishes identically:

$$\pi(t)\alpha + (1-\pi(t))r - \tilde{r}(t) + \mu(t) - \tilde{\mu}(t) = 0.$$

In this case $Y(t)$ is a driftless geometric Brownian motion—an exponential martingale—so the expected payout is constant: $\mathbb{E}[Y(t) \mid Y(m)] = Y(m)$ for all $t \geq m$. This produces a *flat* expected payout profile.

More generally:

- A more conservative (lower) $\tilde{r}(t)$ increases the drift of $Y(t)$, giving an *increasing* expected payout profile (start low, grow over time).
- A more aggressive (higher) $\tilde{r}(t)$ decreases the drift, producing a *decreasing* profile (start high, decline).
- Analogously, choosing $\tilde{\mu}(t) < \mu(t)$ adds positive drift (the basis "under-predicts" mortality), leading to increasing payouts.

The calculation basis thus acts as a lever for shaping the expected trajectory of retirement income.

Exercise (vi)

We simulate $N = 1000$ paths of the life annuity benefit $Y(t)$ for $t \in [m, 111]$ using an Euler scheme with step length $h = 1/100$ on the dynamics derived in Exercise (v):

$$\mathrm{d}Y(t) = \big(\pi(t)\alpha + (1-\pi(t))r - \tilde{r}(t) + \mu(t) - \tilde{\mu}(t)\big)\, Y(t)\,\mathrm{d}t + \sigma\pi(t)\, Y(t)\,\mathrm{d}W(t).$$

With the calculation basis from Table 2 the drift coefficient vanishes, so the Euler step reduces to

$$Y(t_{i+1}) = Y(t_i) + \sigma\,\pi(t_i)\,Y(t_i)\,\sqrt{h}\,Z_i, \qquad Z_i \overset{\text{iid}}{\sim} N(0,1).$$

The initial value is $Y(m) = X(m)/\tilde{V}_0(m)$, where $X(m) \approx 9{,}821{,}181$ is taken from the simulated path in Exercise (ii), giving $Y(70) \approx 827{,}421$. The dashed line in the figure marks this initial benefit level. A selection of 15 sample paths is shown below.

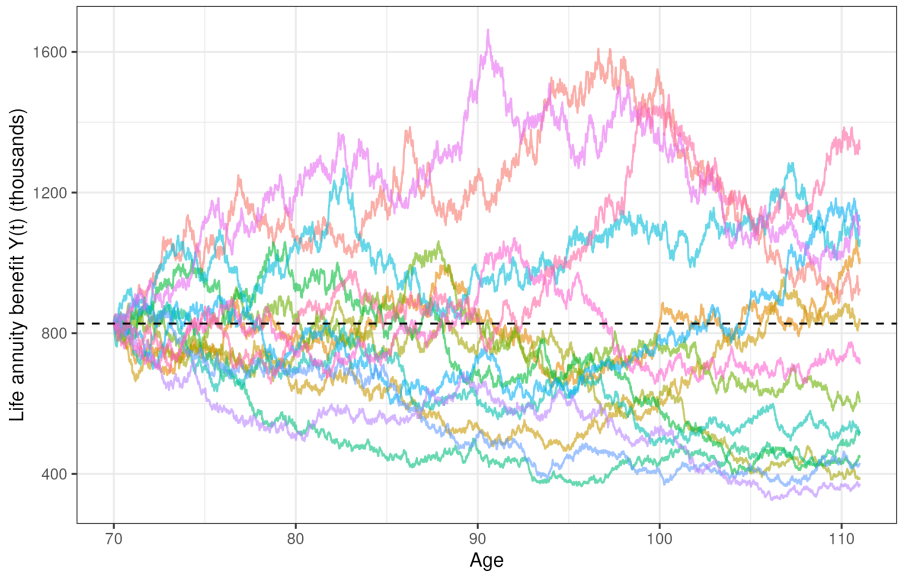


Figure 1: 15 simulated paths of the life annuity benefit $Y(t)$. The dashed line marks the initial benefit $Y(m)$.

We observe significant variation across paths: some paths see benefits double or more, while others decline substantially. This dispersion reflects the stochastic investment risk borne by the insured under the unit-link design.

Exercise (vii)

Based on the $N = 1000$ simulated paths from Exercise (vi), we estimate the expected life annuity benefit $\mathbb{E}[Y(t) \mid X(m)]$ at each time point by the sample mean across paths:

$$\hat{\mathbb{E}}[Y(t) \mid X(m)] = \frac{1}{N} \sum_{n=1}^{N} Y^{(n)}(t).$$

Figure 2: Estimated expected life annuity benefit $\mathbb{E}[Y(t) \mid X(m)]$ over time. The dashed red line marks $Y(m)$.

We observe that the expected payout profile is approximately *flat* over the entire retirement period. This is consistent with the theoretical result from Exercise (v): since $\tilde{r}(t) = \pi(t)\alpha + (1 - \pi(t))r$ and $\tilde{\mu}(t) = \mu(t)$, the drift of Y is zero and Y is an exponential martingale, so $\mathbb{E}[Y(t) \mid X(m)] = Y(m)$ for all $t \geq m$. The small fluctuations visible in the plot are Monte Carlo noise from the finite sample of $N = 1000$ paths.

In practice, this flat expected profile means that, on average, the insured receives the same benefit throughout retirement. However, as seen in Exercise (vi), individual paths exhibit substantial variability around this expectation—the insured bears significant investment risk. A more conservative calculation basis (with $\tilde{r}(t) < \pi(t)\alpha + (1 - \pi(t))r$) would instead produce an increasing expected payout profile, starting lower but growing over time, which could be preferable from a product design perspective.